

Effects of Dual Tasks on 30-Minute Treadmill Walking: Pilot Test

YH Cho^{1,4}, I JO^{1,4}, YE Choi^{1,4}, JW Kim¹, SJ HAN¹, TH KIM³, Hae-Dong Lee^{2,4*}

¹Graduate School of Yonsei University, Seoul, Korea;

²Department of Physical Education, College of Educational Sciences, Yonsei University, Seoul, Korea;

³Department of Sport Industry Studies, Yonsei University, Seoul, Korea

⁴Frontier Research Institute of Convergence Sports Science, College of Educational Sciences, Yonsei University, Seoul, Korea

INTRODUCTION

Elderly – walking

- Characteristic

Walking speed ↓
Step length ↓
Step width ↓
Gait variability ↑

- Fall risk

- Evaluation tool

Functional
Health outcomes

(Borst et al., 2008, Van Kan et al., 2009, Öberg et al., 1993, Callisaya et al., 2011)



INTRODUCTION

Elderly – treadmill

Initial phase

Lower limb joint angle ROM ↑
Stride time variability ↑

30-min walk

Stride length variability ↓
Stability of gait ↑



INTRODUCTION

Dual task

Music

Stride cadence ↑

Positive effects

synchronized with
walking rhythm

Video

(sound, subtitles)

stride interval
variability ↑



Research purpose

- **Effect of video-watching as a dual-task during 30-minute treadmill walking on gait variability in older adults**

METHODS

- 10 female elderly

Age: 70.9 ± 4.0 y

Height: 155.6 ± 4.6 cm

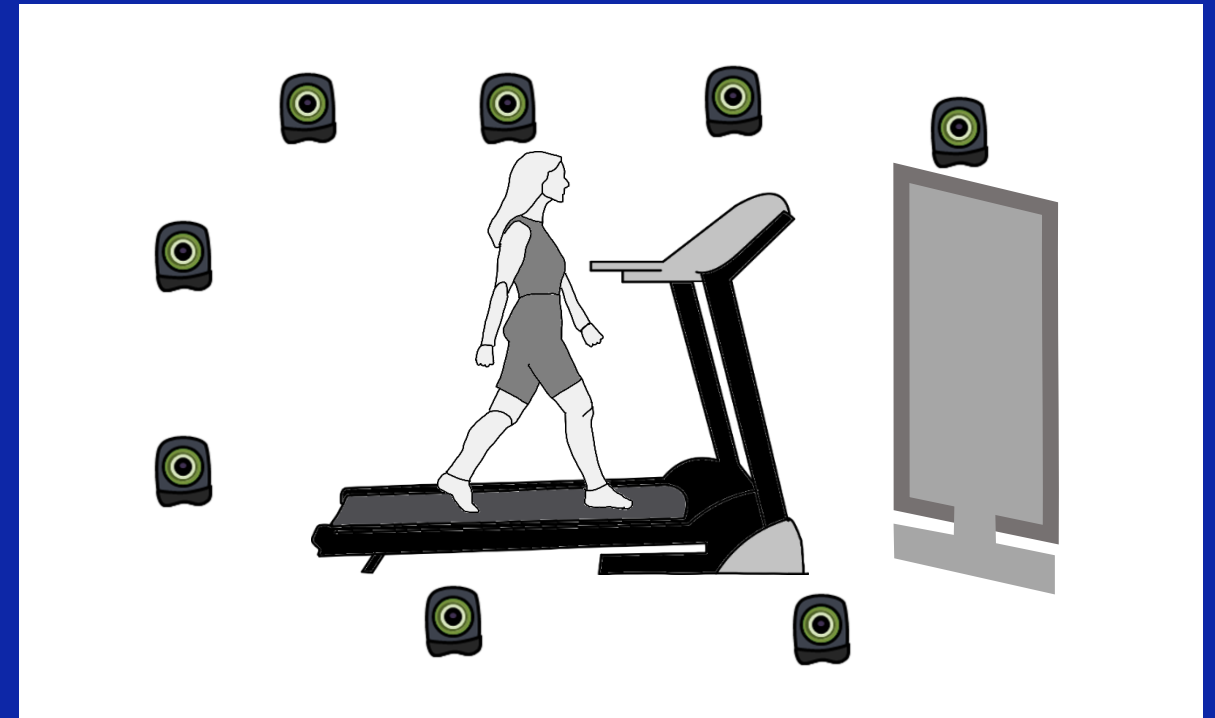
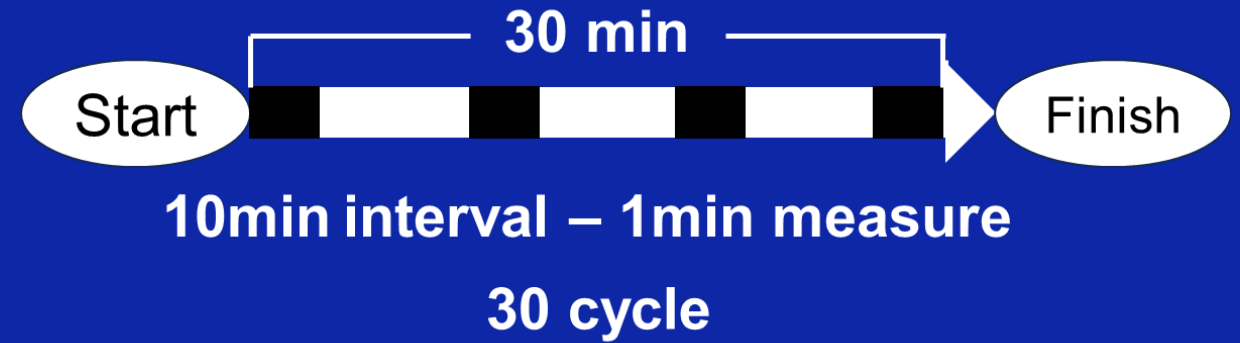
Weight: 56.3 ± 6.5 kg

Preferred speed: 1.8 ± 1.2 km/h

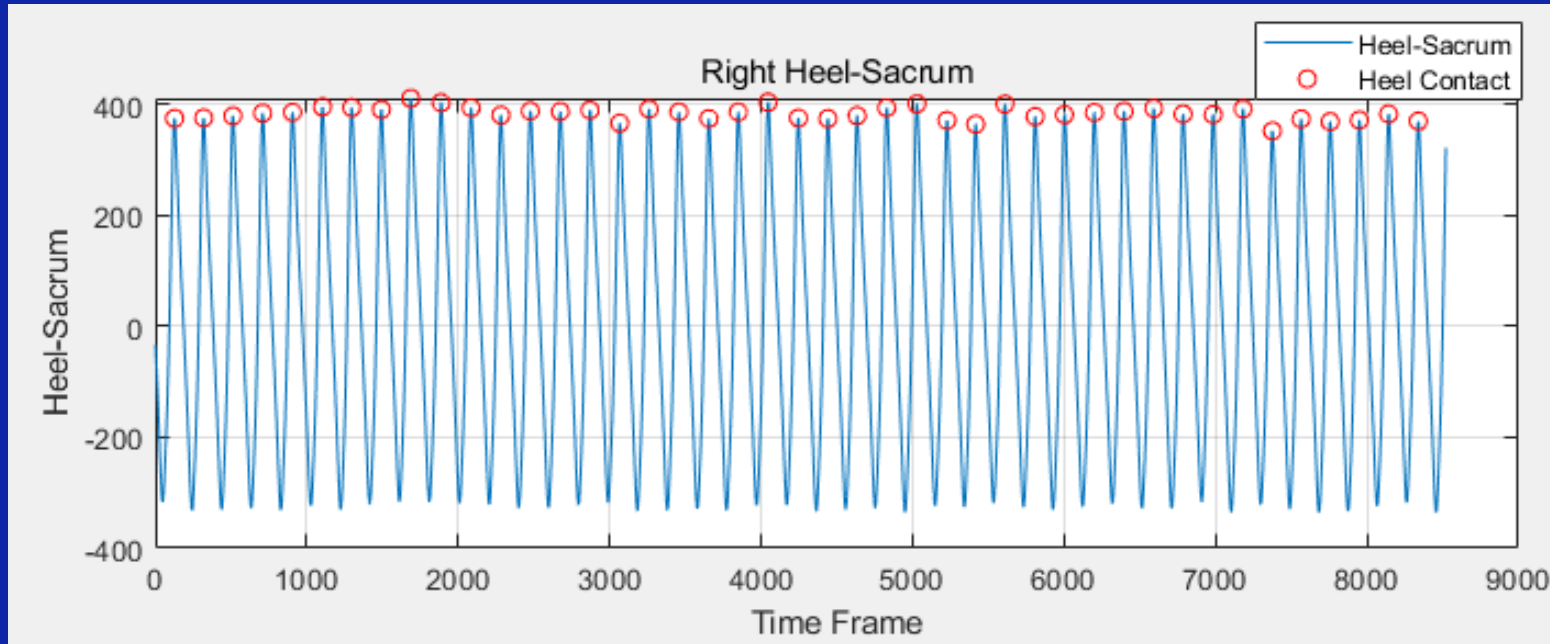
No experience of falling (~1year)

Cognitive Impairment Screening Test (CIST)

Randomized order (single vs. dual task)
(At least 24h after)

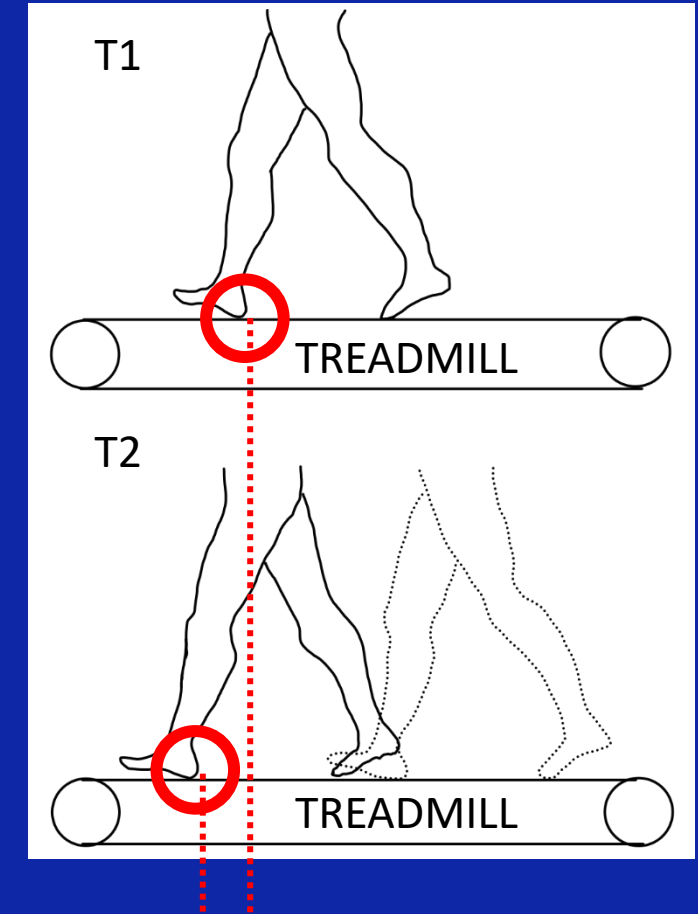


METHODS



$$\text{Heel strike} = (X_{\text{heel}} - X_{\text{sacrum}})_{\text{max}}$$

$$\text{Toe off} = (X_{\text{off}} - X_{\text{sacrum}})_{\text{min}}$$



$$\text{Step length} = (T2 - T1) \times V + (X2 - X1)$$

$$\text{Step width} = Y2 - Y1$$

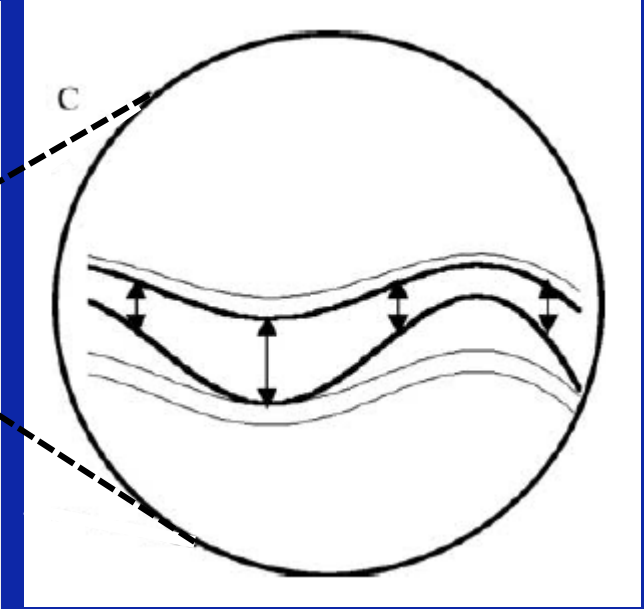
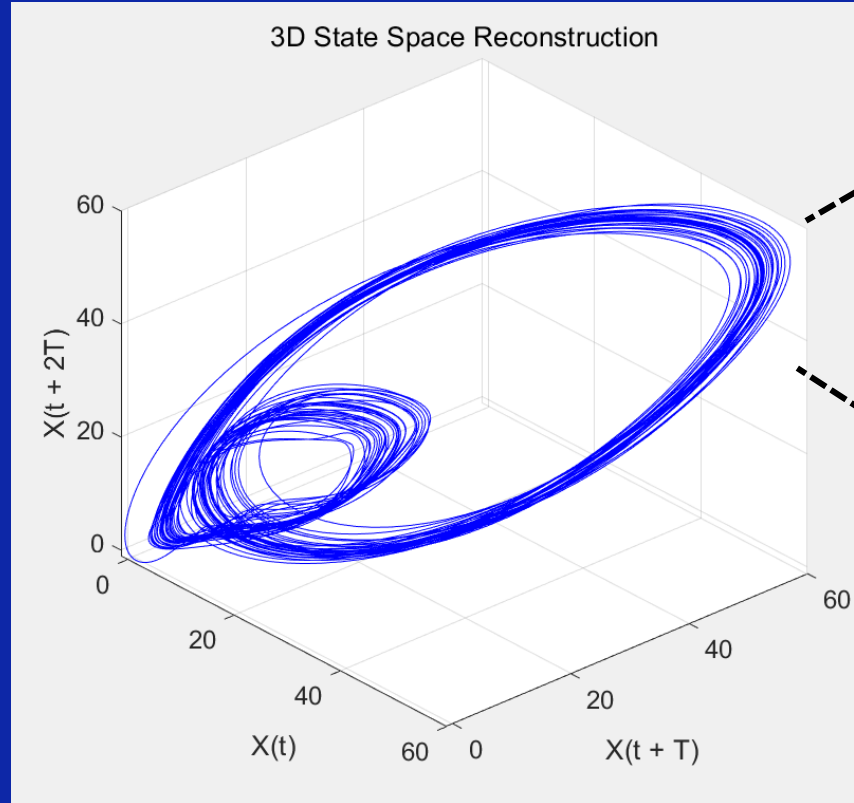
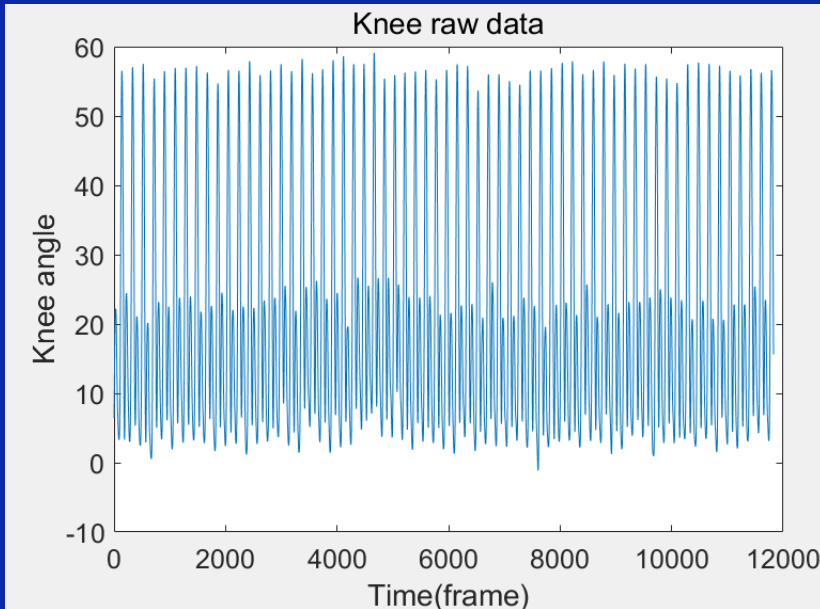
X1 : Heel strike T1

X2 : Heel strike T2

METHODS

Lyapunov Exponent(LyE)

- Nonlinear dynamics



Variability
(Coefficient of Variation)

$$X(t) = [x(t), x(t+T), x(t+2T), \dots, x(t+(d-1)T)] \quad (1)$$

$$d(t) = d_0 e^{\lambda_1 t} \quad (2)$$

$$\ln[d_j(i)] \approx \lambda^*(i\Delta t) + \ln[d_{0j}] \quad (3)$$

$$y(i) = \frac{1}{\Delta t} < \ln[d_j(i)] > \quad (4)$$

$$= \frac{SD(\text{standard deviation})}{Mean} \times 100\%$$

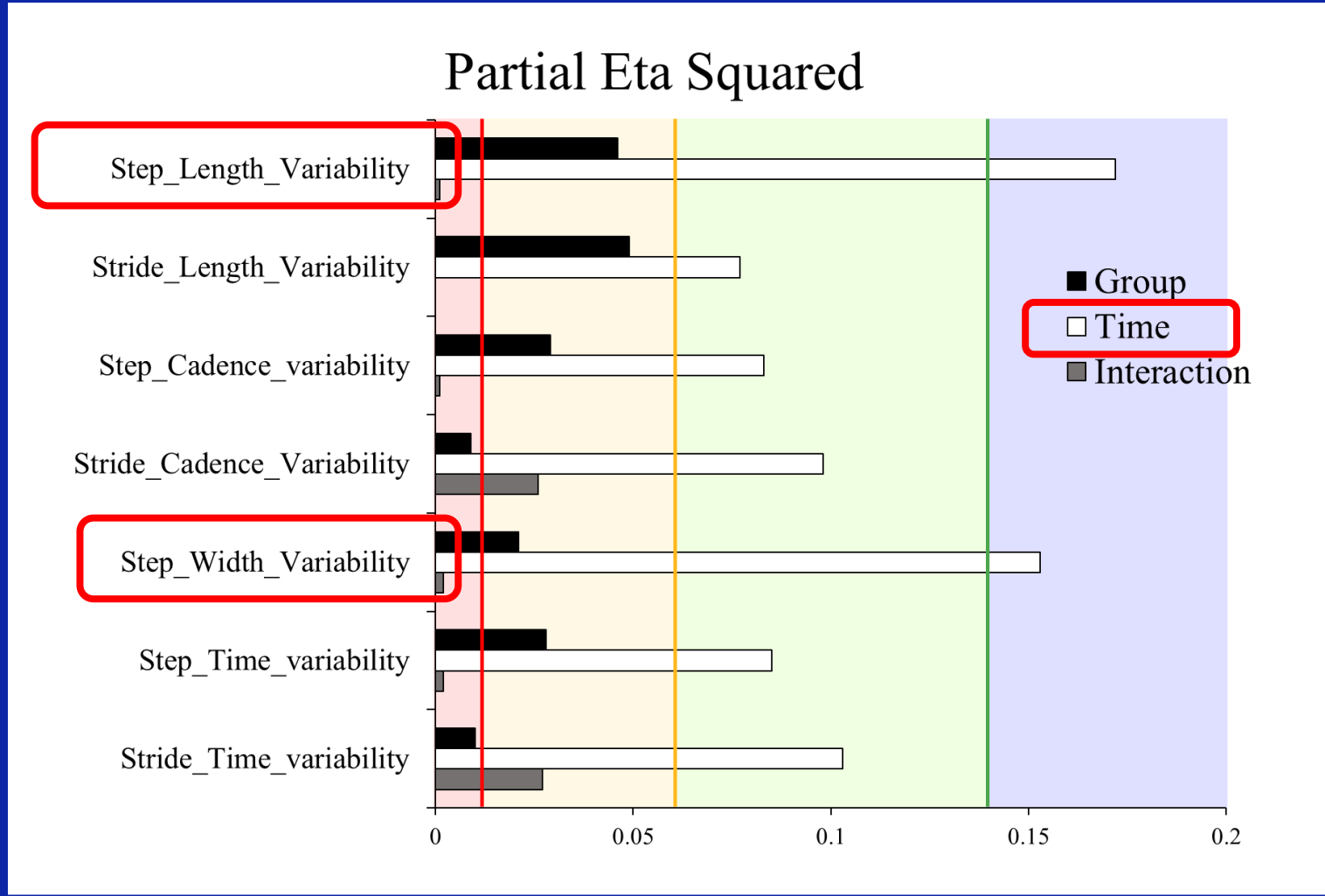
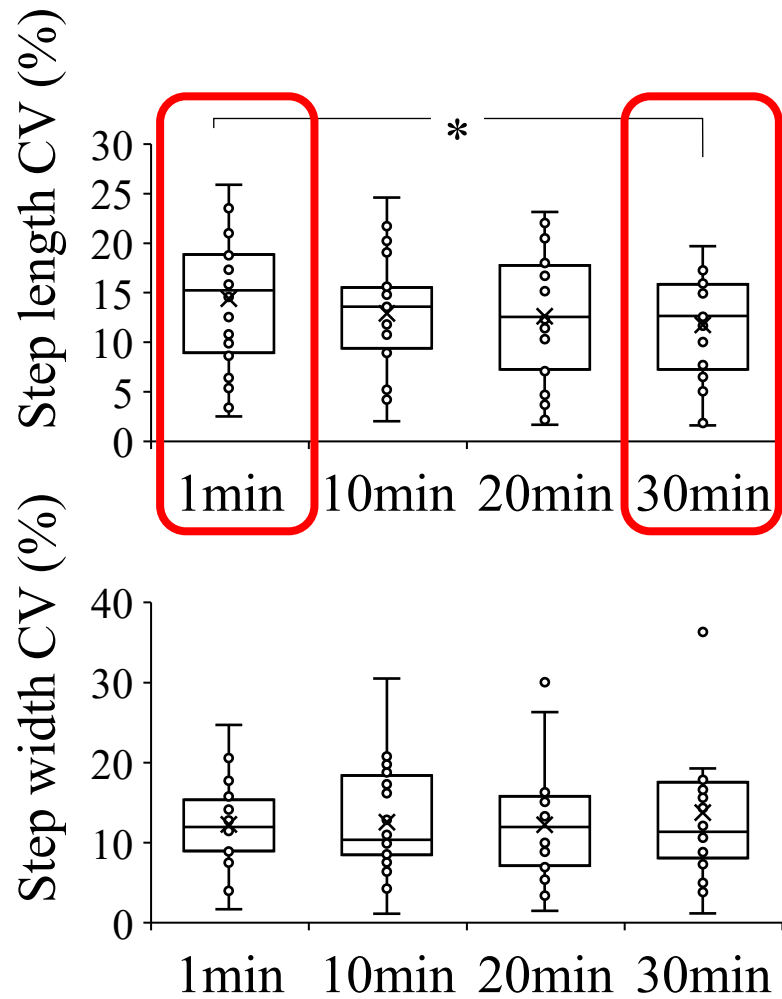
METHODS

Two-way Repeated Measures ANOVA Partial Eta Squared(effect size)

Task x Time
(single vs dual) (1m, 10m, 20m, 30m)

Gait variables (e.g., step length, step variability, etc.)
and joint variability

Results & Discussion

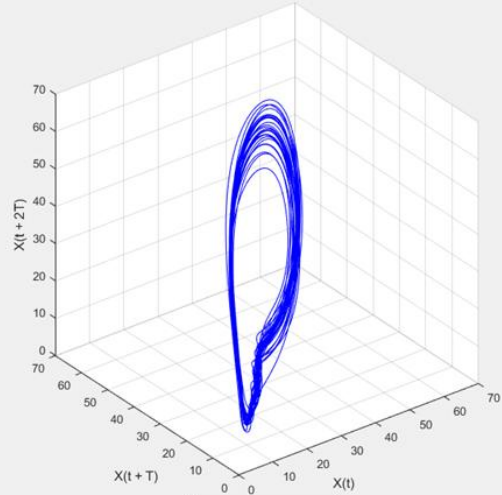


Results & Discussion

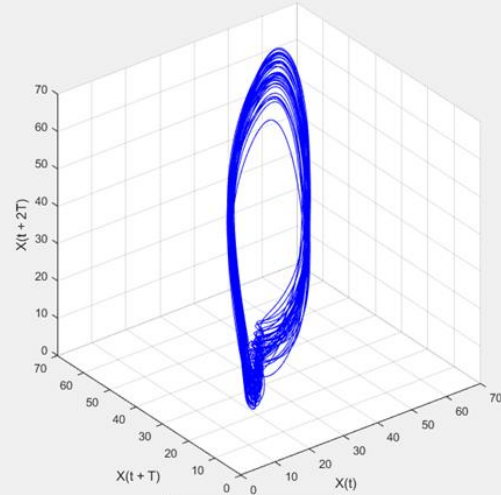
Subject 2

Knee angle LyE (Flex/Ext)

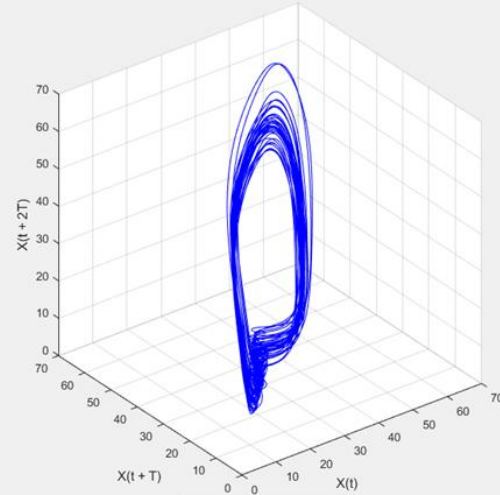
Single



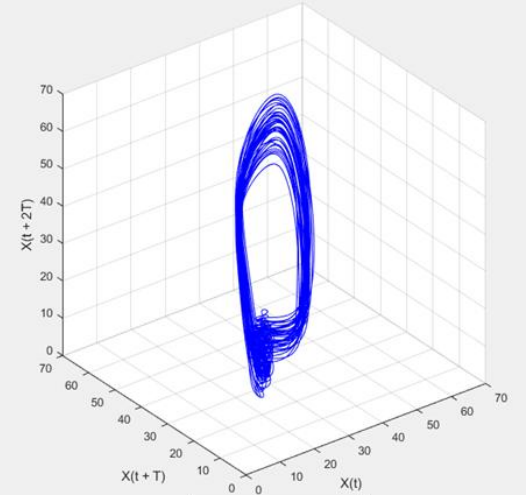
1min



10min

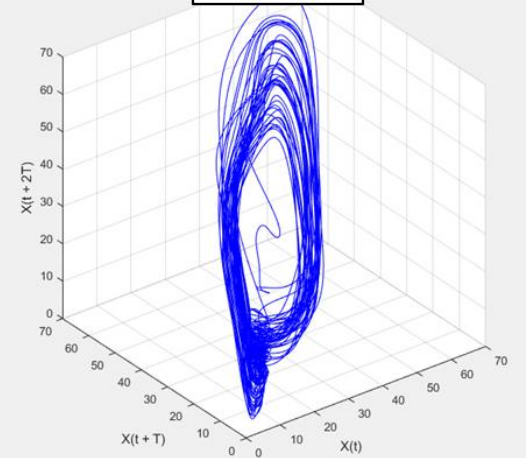
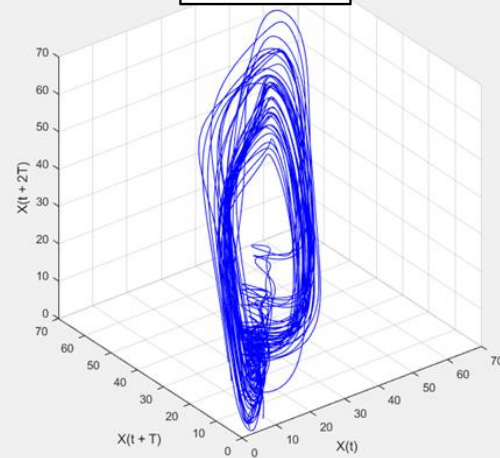
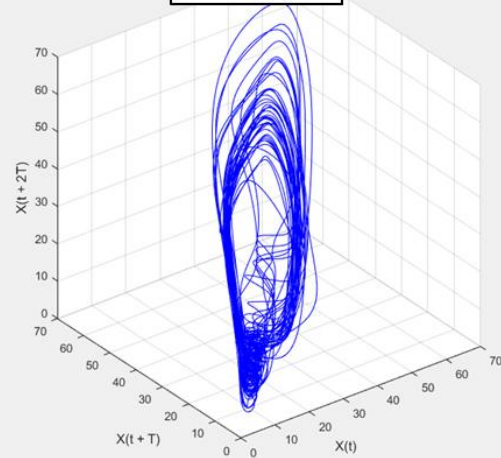
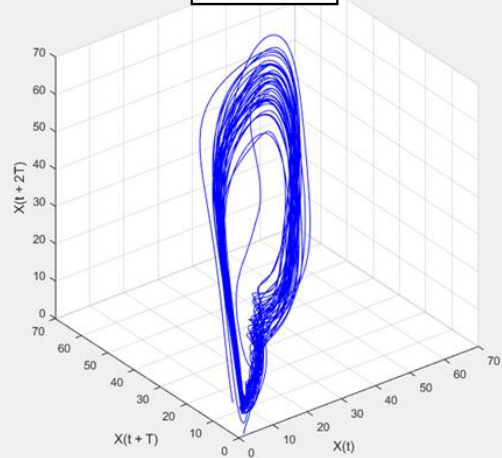


20min



30min

Dual

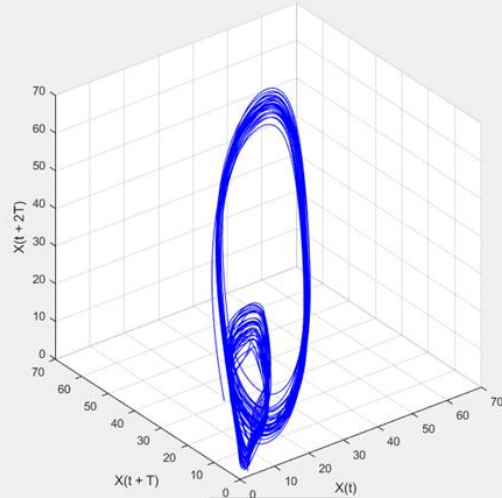


Results & Discussion

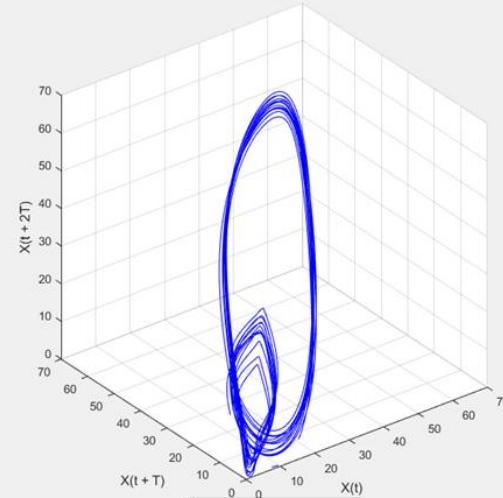
Subject 5

Knee angle LyE (Flex/Ext)

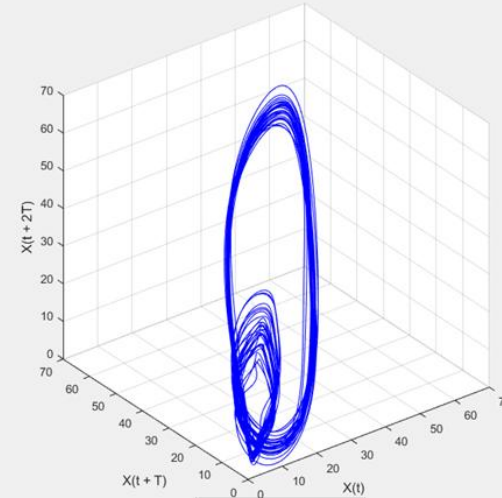
Single



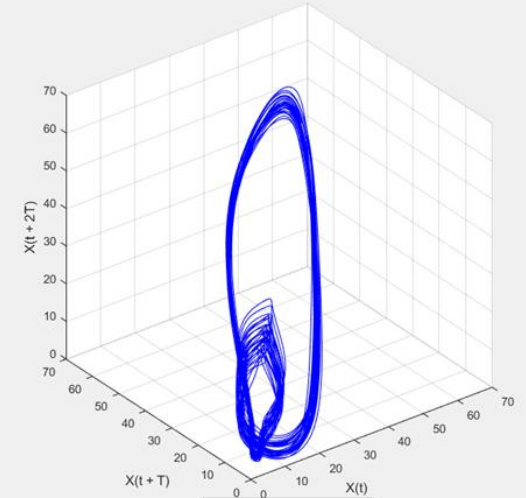
1min



10min

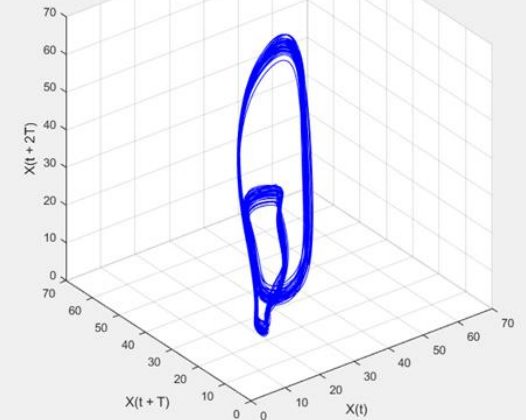
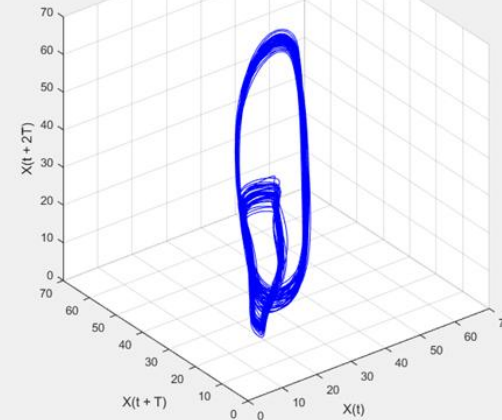
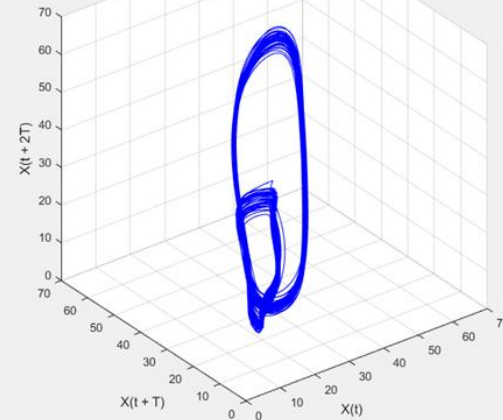
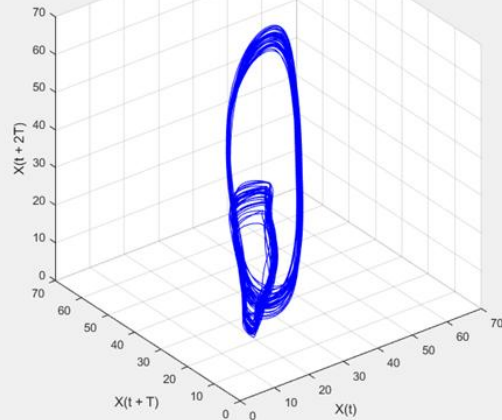


20min

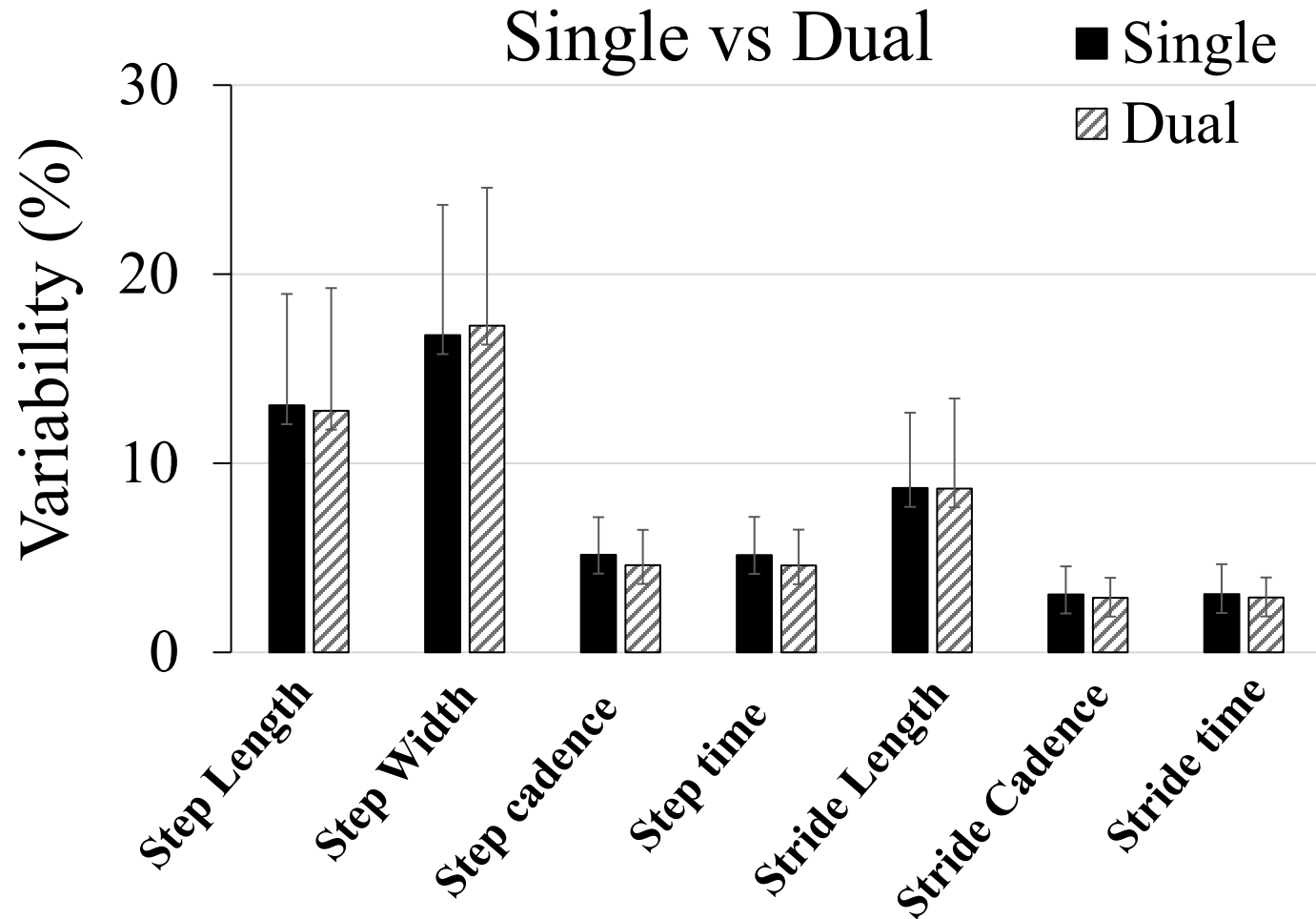


30min

Dual



Results & Discussion



Treadmill speed

Walking control

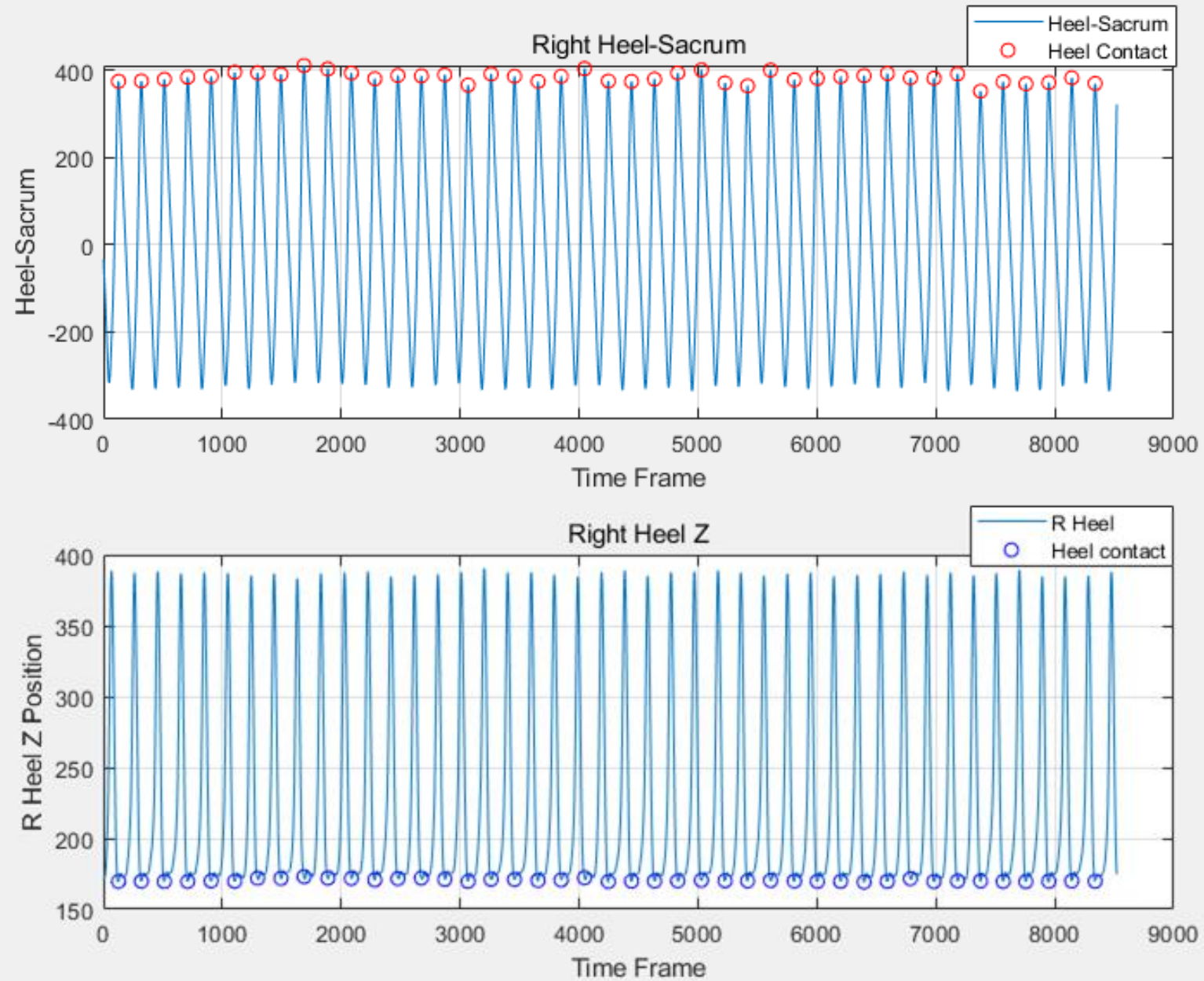
Conclusion

- **Step Length Variability was high at the beginning of treadmill walking, but the variability decreased over time.**
- **Watching video with sound did not induce differences in gait posture control compared to normal walking.**

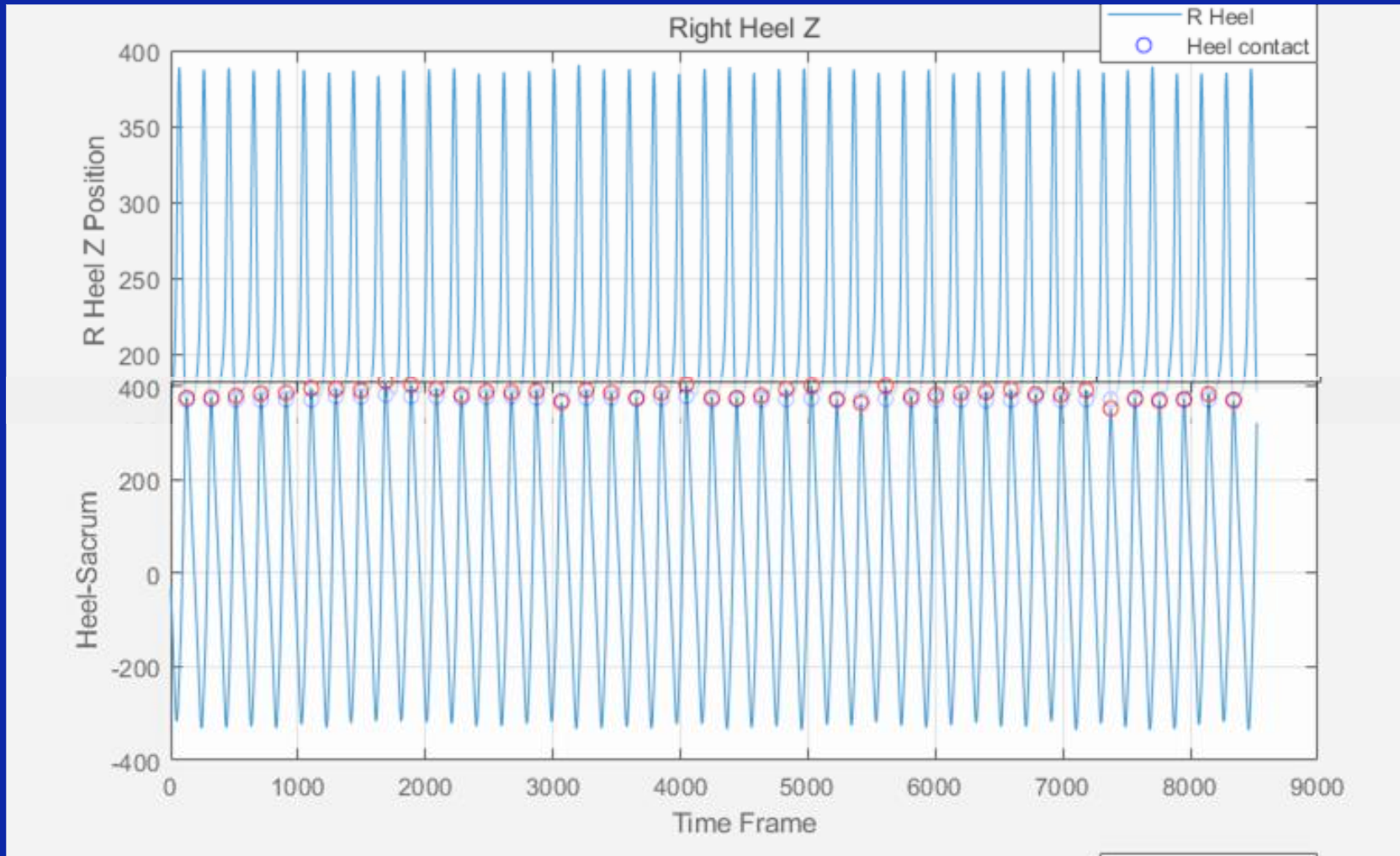
Thank you 🤗

Q&A

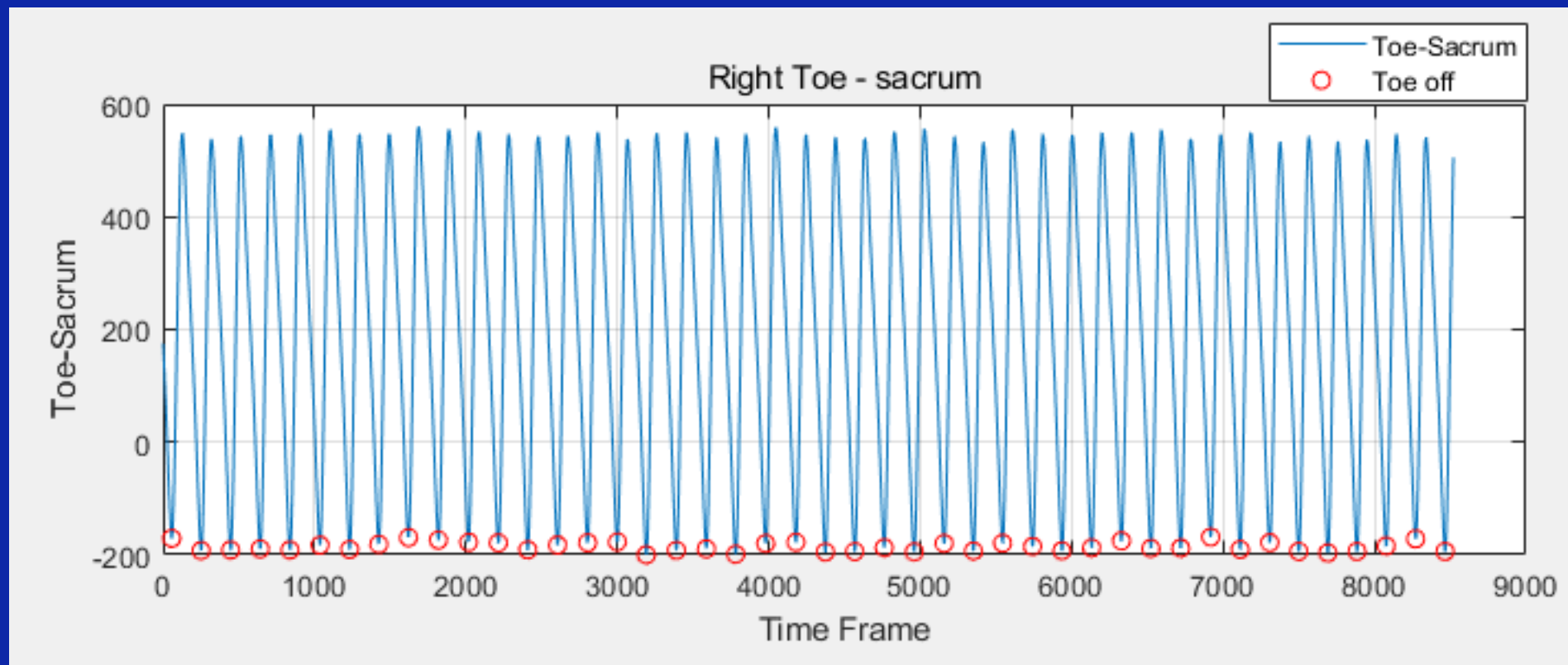
Q&A



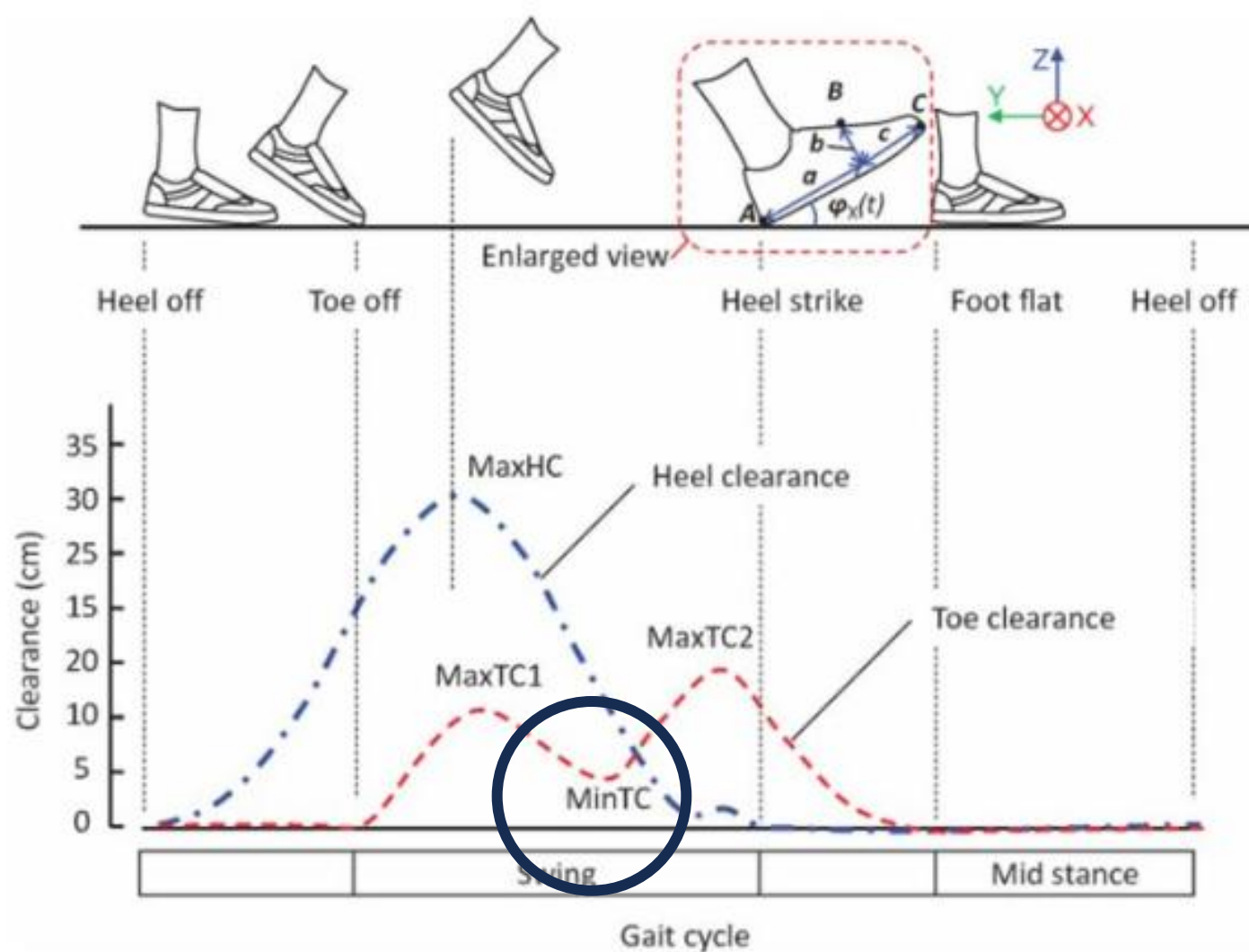
Q&A



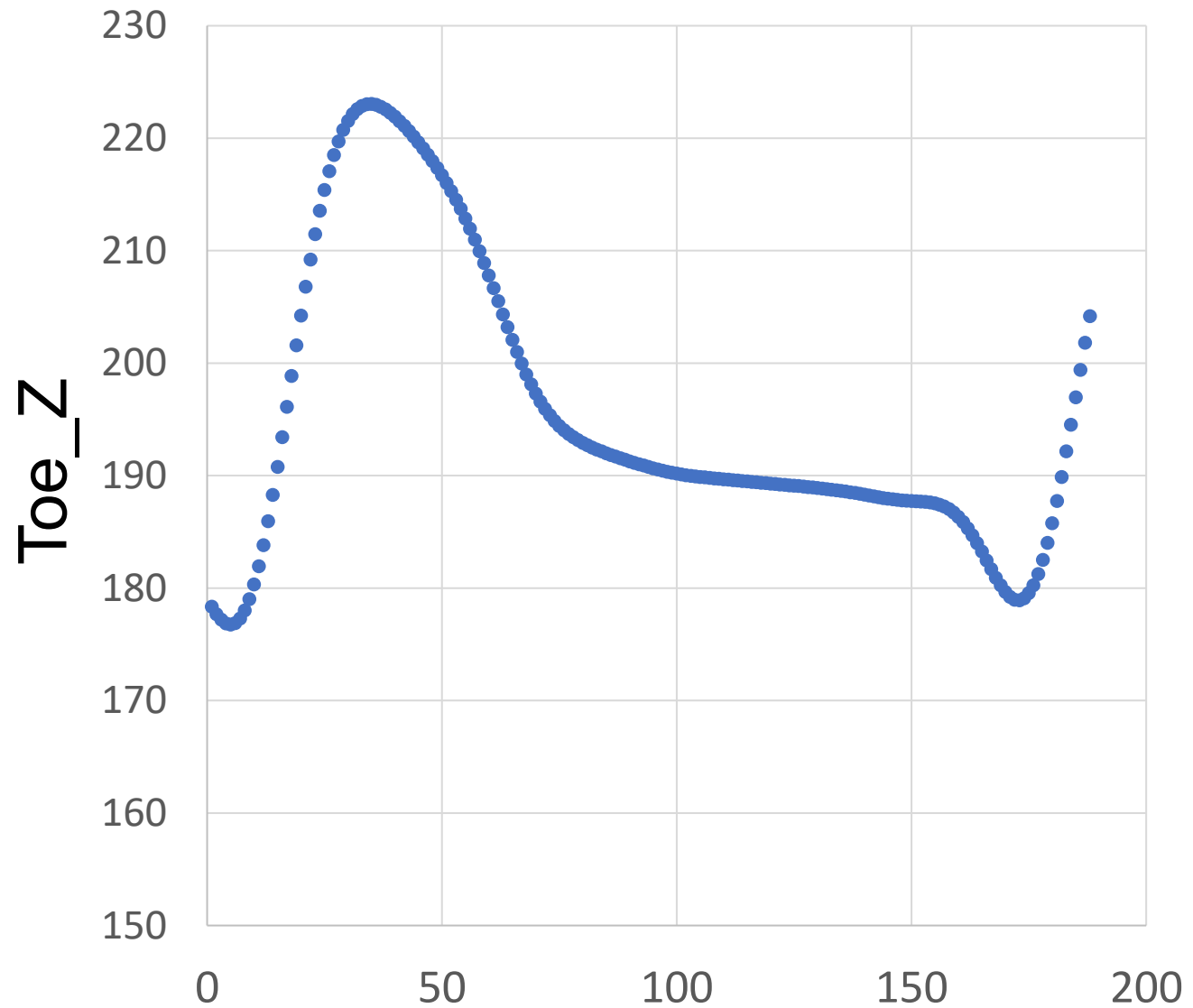
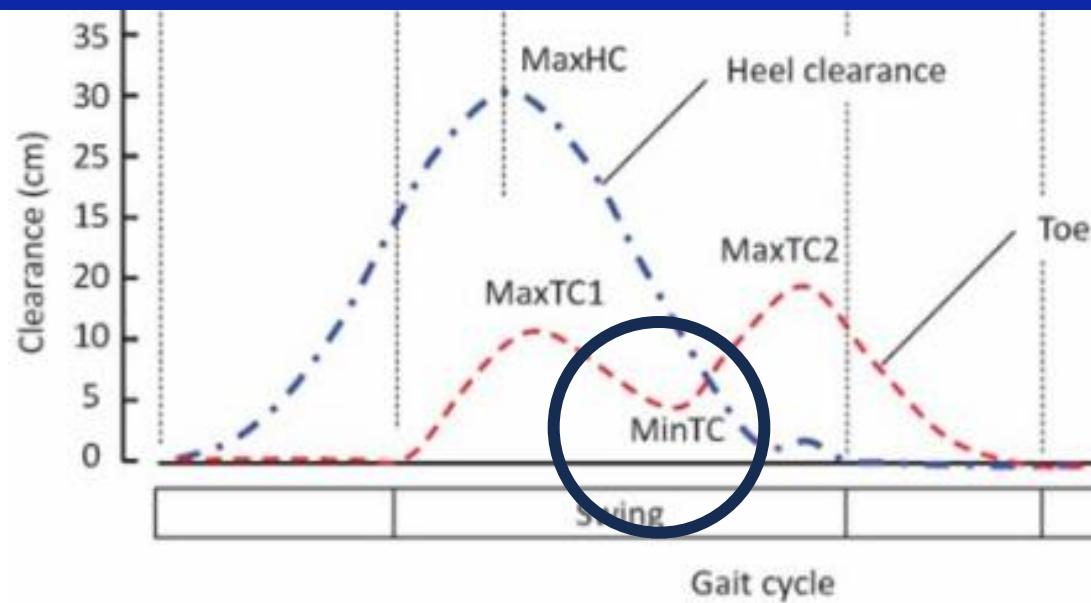
Q&A



Q&A

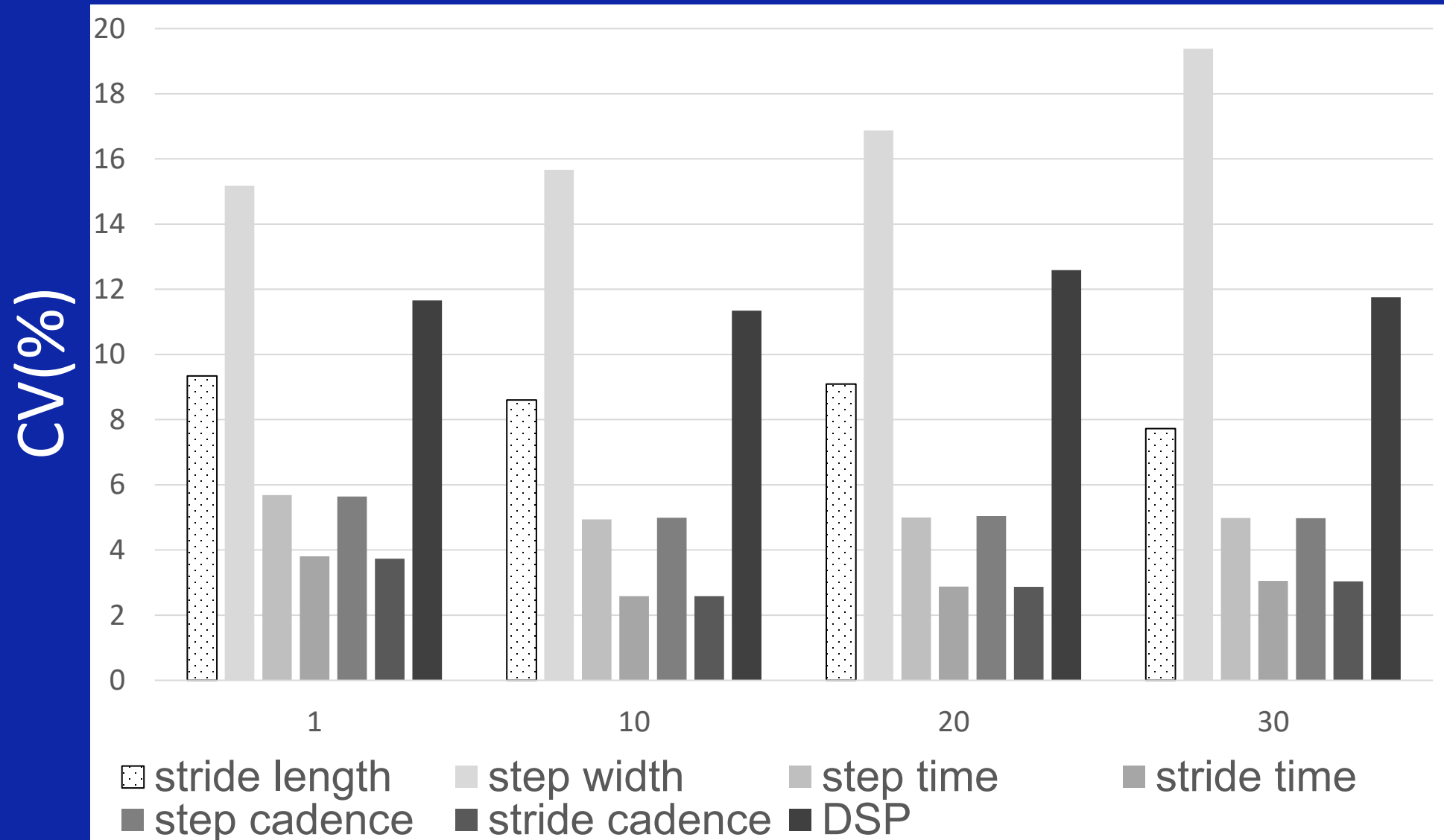


Q&A



Q&A

Single task



Q&A

Dual task

